28/02/23

1. a) $4.18 \times 50 \times 43 = 8987 = q$
   b) $4.18 \times 125 \times 16 = 8360 = q$

2. $\frac{1.29}{86} = 0.015$

$$\frac{-q}{n} = \frac{-34.2}{0.015} = -2280\ kJ/mol$$

3. $C_2H_5OH + 2O_2 \rightarrow 3H_2O + CO_2$

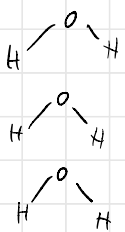
C = 2	C = 1
H = 6	H = 2 × 6
O = 2 × 5	O = 3 × 5

$2H_{2(g)} + O_{2(g)} \rightarrow 2H_2O_{(g)}$

endothermic — energy in

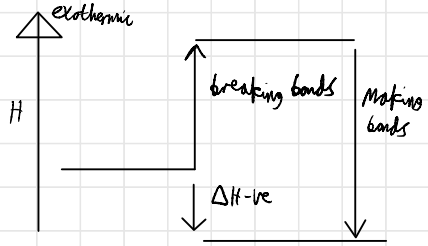
exothermic — energy out

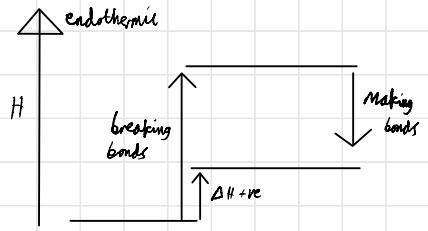
H — H
H — H
O = O



endothermic reaction → energy needed to break bonds > energy needed to make bonds

exothermic reaction → energy needed to make bonds < energy needed to make bonds





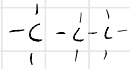
Key facts about bond enthalpies

- energy always required to break bonds.
- energy required to break bonds has same value as energy required to make the bond.
- Bond enthalpies always endothermic
- Always have positive enthalpy value

$\Delta H = \Sigma$ (bond enthalpies in reactants) $- \Sigma$ (bond enthalpies in products)

eg: $436 \times 2 + 498 - (-464 \times 4) = 3226$

1. $C_3H_{8(g)} + 5O_{2(g)} \rightarrow 3CO_{2(g)} + 4H_2O_{(g)}$



$8(C-H) = 413 \times 8 =$
$2(C-C) = 347 \times 2 =$ +
3998

$5(O=O) = 5 \times 498$ +
6488

Enthalpy change of reaction: Standard Condition, Standard States.

Average enthalpy = not standard conditions, gaseous.

03.03.23

1. average bond enthalpy = average amount of energy to break one mole of a bond.
2. Not accurate. Different energy required in different molecules.
   may not be standard because gaseous.

$\Delta_r H$ → does n have to be balanced to 1 mole
↳ Applies to all types of reaction

$\Delta_f H$ C_4H_{10} → compound
C H

Enthalpy of formation of an element is always 0. No energy change.

$\Delta_{neut} H$ → value always the same no matter the reactants. 1 mole of water formed.
$H^+ + OH^- \rightarrow H_2O$

$q = mc\Delta t$ = energy change = heat energy in J.
$H = kJ/mol = \frac{-q}{n}$ = enthalpy.

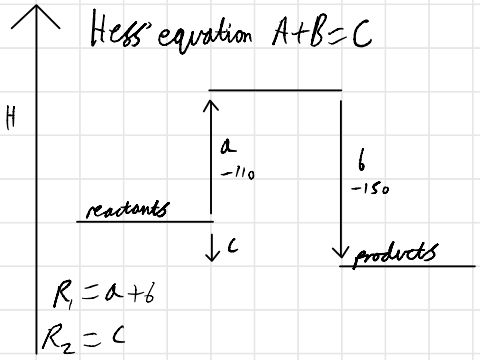
How to determine enthalpy change

1. experimentally
   $q = mc\Delta t$ mass water $1 cm^3 = 1g$
   $n = \frac{m}{M}$ or $n = \frac{CxV}{1,000}$
   $\Delta H = \frac{q}{n}$

   $\Delta_r H^\theta$ Solution + Solid
   $\Delta_c H^\theta$ ← standard = 2 solids
   $\Delta_{neut} H^\theta$ = mass of water = Σ of solutions

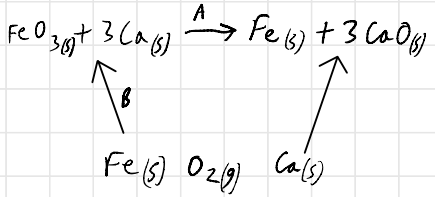
2. average bond enthalpy data
   $\Delta H = \Sigma$ reactants $- \Sigma$ products

3. Hess' Law
   If reaction can take place by two routes and starting and finishing conditions are same - total enthalpy is same.



Fe_2O_3 → -824 Fe 0

Ca 0 CaO -635



R_1: B + A

R_2: C

$B + A = -824 + 3(0) + A$

$FeO_3 = -824$

$Ca = 0$

$C = 3(-635)$

$-824 + A = -1905$

$A = -1905 + 824$

$A = -1081\ kjmol^{-1}$

$\Delta_f H$ elements → products

$\Delta_c H$ reactants → combustion products

Page 139

$2(-46) - 93 = -139$

$-314 - -139 = -175$

b) $2(-314) - 986 + B = -796 + 2(-48) + 2(-286)$

$B - 1614 = -1460$

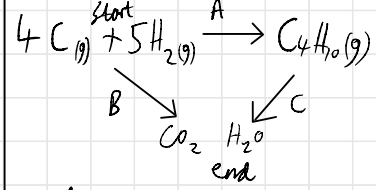
$B = 154$

c) $2(-48) + 3(-157) + B = 0 + 3(0) + 3(-286)$

$B - 563 = -858$

$B = -295$

Enthalpy of combustion



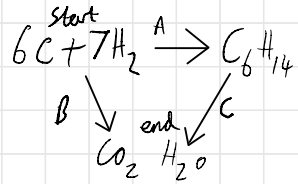
R_1: A + C

R_2: B

$HC = A + C = B$

$(4x-394 + 5x-286) - -2877 = -129$

a) C_6H_{14}



$B = A + C$

$(6x-394 + 7x-286) = A + -4163$

$A = -203$

07.03.23

8. a) enthalpy change of combustion = amount of energy to react with 1 mol of O_2.

b) $q = mc\Delta t$

$250 \times 4.18 \times (78-24) = 56430\,J$

$56.430\,kJ$

$n = \frac{m}{M}$ $\frac{1.76}{CH_3(CH_2)_4OH} =$ $12+3+(12+2)\times4+16+1 = 88$ $\frac{1.76}{88}$

$\Delta H = \frac{-56.43}{0.02} = -2821.5\,kJ/mol$

c) look pg 29 8k

$6C_{(s)} + 7H_{2(g)} \rightarrow C_6H_{14(l)}$

Σ bonds broken − Σ bonds made = average bond enthalpies

reactants

$6C + 7H_2 \xrightarrow{A} C_6H_{14}$

$\downarrow B$ $\quad$ $\swarrow C$ -4166

$H_2O + CO_2$

$A + C = B$ $\quad (-394 \times 6) + (-286 \times 7)$

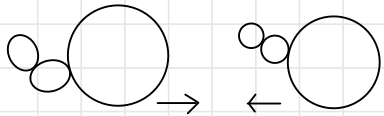
$B - C = A$

$-4366 - -4166 = -200$

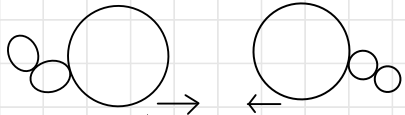
Collision theory

for effective collisions:

1. particles collide with correct orientation.
2. particles need enough energy to overcome activation energy barrier for reaction.



No reaction



reaction takes place

$\text{rate} = \frac{\text{change in conc}}{\text{time}}$ $\quad mol\,dm^{-3}\,s^{-1}$

Factors to affect rate of reaction

- Concentration (pressure)
- Temperature
- Use of Catalysts
- Surface area of solid reactants

Modelling Concentration

- Particles closer
- More freq collisions
- More successful collisions
- Increase conc = increase rate

10.03.23

Activation energy = minimum energy to start reaction by breaking bonds

Transition State = unstable, highest point on curve
activation energy = enthalpy difference between reactants and transition state

Catalyst = alternative energy pathway, not used up.

Pros:	Cons:
Cheaper	Some catalysts are toxic
reusable	
quicker reaction	
less fossil fuels burned	
less CO_2 released	

Heterogenous Catalysts

↓

catalysts in different state to reactants.
eg: Haber Process

1. weak bonds made with surface of catalyst = adsorption
2. reaction takes place, products formed
3. products released from catalyst = desorption

Process	Catalyst	Equation
Making ammonia	$Fe(s)$	$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$
reforming	$Pt(s)$ or $Rh(s)$	$C_6H_{14(g)} \rightarrow C_6H_{12(g)} + H_{2(g)}$
hydrogenation of alkenes	$Ni(s)$	$C_2H_{4(g)} + H_{2(g)} \rightarrow C_2H_{6(g)}$
making sulfur trioxide for Sulfuric acid (contact process)	$V_2O_5(s)$	$2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}$

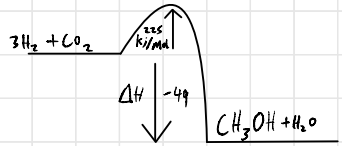
Homogeneous catalyst

$$2O_{3(g)} \overset{Cl_{(g)}}{\rightleftharpoons} 3O_{2(g)}$$

$$Cl^{\cdot} + O_3 \rightarrow ClO^{\cdot} + O_2$$
$$ClO^{\cdot} + O_3 \rightarrow Cl^{\cdot} + 2O_2$$

$$C_2H_5OH(l) + CH_3COOH(l) \overset{H_2SO_4(l)}{\rightleftharpoons} CH_3COOC_2H_5(l) + H_2O(l)$$

1. Homogeneous are same state as reactants.
2. lowers activation energy and decreases overall enthalpy change.
3. a)



b) $49 + 225 = 274\ kJmol^{-1}$

21.03.23

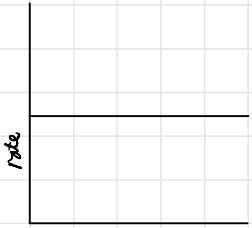
[A] = Concentration of A

rate of reaction is proportional to concentration raised to a power.

Rate $\propto [A]^n$

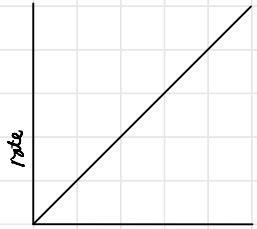
n = order of reaction

Zero order $[A]^0$, First order $[A]^1$, Second order $[A]^2$



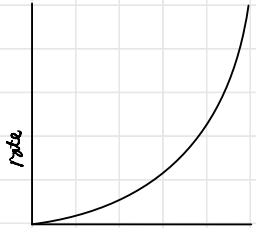
rate $\propto [A]^0$

Zero order = concentration has no effect on rate



rate $\propto [A]^1$

1st order = concentration is directly proportional to rate



rate $\propto [A]^2$

2nd order = Concentration Squared = rate

2. a) i) nothing ii) doubles iii) quadruples

b) x8

Overall Rate

$rate = k[A]^m[B]^n \quad 0 \leq m \leq 2, \quad 0 \leq n \leq 2$

Eg. $A + B + C \rightarrow$ Products

$rate = k[A]^0[B]^1[C]^2$

$rate = kBC^2$

1. $[A]^1[B]^2[C]^1 k$

kAB^2C

24.03.23

3. a) A¹, B⁰, C¹

b) 2

c) rate = k [A][C]

$mol\,dm^{-3}\,s^{-1} = mol\,dm^{-3} \times mol\,dm^{-3}$

0 order reaction

rate = k [A]⁰

= k × 1

= k

$mol\,dm^{-3}\,s^{-1}$ = units for k

1st order reaction

rate = k [A]¹

rate = kA

$k = \frac{rate}{A}$

$k = \frac{mol\,dm^{-3}\,s^{-1}}{mol\,dm^{-3}}$

$k = s^{-1}$

2nd order reaction

$rate = k[A]^2$

$rate = A^2 k$

$k = \frac{rate}{A^2}$

$k = \frac{mol\,dm^{-3}\,s^{-1}}{(mol\,dm^{-3})^2} = \frac{s^{-1}}{mol\,dm^{-3}}$

$= s^{-1}\,mol^{-1}\,dm^3$

$\frac{mol\,dm_1^{-3}\,s^{-1}}{mol^2\,dm^{-6}\;\; mol^2\,dm^{-6}}$

$\frac{mol\,dm^{-3}\,s^{-1}}{mol^4\,dm^{-12}}$ $\quad mol^{-3}\,dm^9\,s^{-1}$

$CuSO_4 \cdot 5H_2O \rightleftharpoons CuSO_4 + 5H_2O$

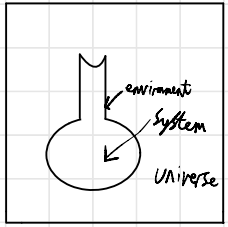
$A + B \rightleftharpoons C$

At dynamic equilibrium:

- A and B react to make C
- Simultaneously, at the same rate C reacts to make A and B

.

.



Le Chatelier's Principle = When a system in equilibrium is subjected to an external change to the system re-adjusts itself to minimise the effect of change.

Exothermic Reactions

Chemical energy → Heat energy

$$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$$

exothermic →

← endothermic

K_c - the equilibrium constant

- Gives position of equilibrium
- Magnitude indicates whether more reactants or products

$$A + B \rightleftharpoons C + 2D$$

$$K_c = \frac{[C][D]^2}{[A][B]}$$

1. $\frac{[NH_3]^2}{[N_2][H_2]^3}$

$$K_c = \frac{[2.28 \times 10^{-3}]^2}{[N_2][H_2]^3}$$

$$\frac{(2.28 \times 10^{-3})^2}{(8 \times 10^{-4})(1 \times 10^{-4})^3} = 6,498,000$$

1. $K_c = \frac{0.25 \times 0.4}{0.25 \times 0.1} = 4$

4 > 1, equilibrium shifts to Products

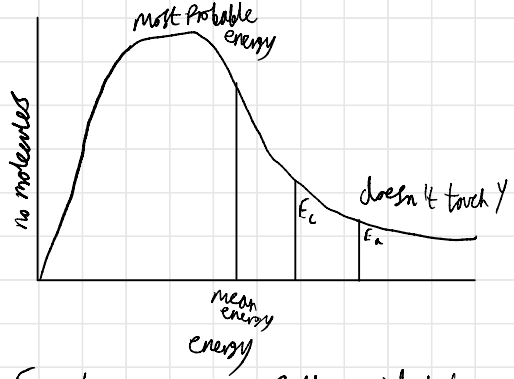
2. $\frac{0.4 \times 0.3}{(0.14)^2} = 6.12244898$

3. $\frac{(0.23)^2 \times 1.37}{(0.93)^2} = 0.1173$

2 8 0 3 2 3

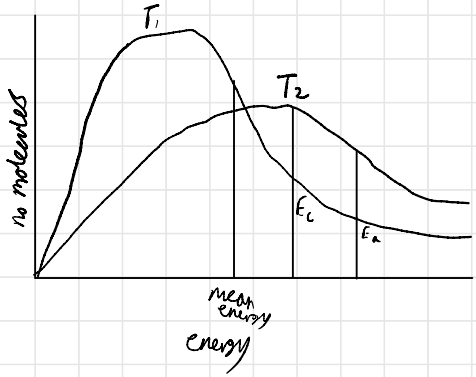
Covalent bonding = Strong electrostatic attraction between shared pair of electrons.

Boltzmann distribution



E_c = lower energy path provided by catalyst.

- No molecules have zero energy
- area under curve = total number of molecules
- no maximum energy for molecule, infinite energy to touch x axis.

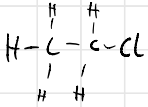


At higher temperature:

- more molecules have energy greater or equal to activation energy
- greater proportion of effective collisions
- More frequent collisions because molecules are moving faster

$$K_c = \frac{[products]}{[reactants]}$$

18.04.23

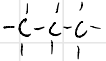
1. 
Chlorine

2. Homologous Series = Same chemical properties, Successive member differs by CH_2

3. Aromatic, Aliphatic, Alicyclic

4. 2-3 dichloropentane

Stem → number of carbon on longest chain
Suffix → ester
Prefix → before

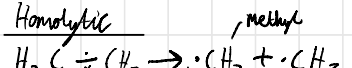
Types of formula
- molecular = C_3H_8
- Empirical = C_3H_8
- Structural = $CH_3CH_2CH_3$
- Skeletal =
- Displayed = 
- General = C_nH_{2n+2}

1. C_4H_{10}

Saturated = Single C-C bonds only
Unsaturated = multiple C bonds, double or triple

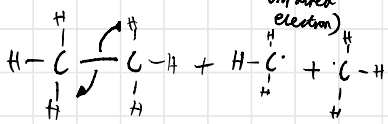
Homolytic fission = each bonded atom takes 1 of the shared pair of e⁻
Heterolytic = 1 bonded atom takes both shared pair of e⁻
covalent bond = shared pair of e⁻

Homolytic



Chemical equation

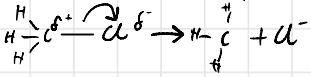
reaction mechanism



Heterolytic

$H_3C-Cl \rightarrow CH_3^+ + Cl^-$ = Chemical equation

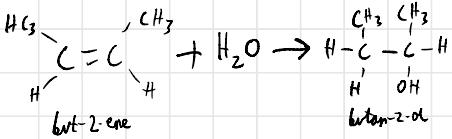
reaction mechanism



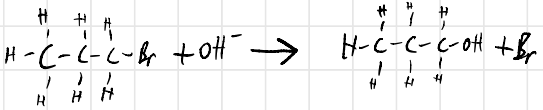
Homolytic forms radicals
Heterolytic forms ions

Types of reaction:
1. Addition = 2 reactants form a product
2. Substitution = atom is replaced by different atom
3. Elimination = removal of small molecule from larger molecule.

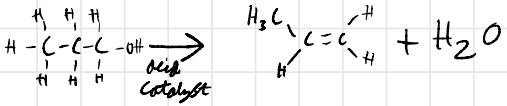
Addition



Substitution



Elimination



1. Heterolytic fission = 1 donated takes both pair of e⁻
2. radical = alkyl + electron
3. Sub
   Add
   Elim
   Add

Properties of Alkenes

Starter

1. homolytic fission = each bonded atom takes 1e⁻ each ✓
2. heterolytic fission = 1 bonded atom takes both of the pair. ✓
3. Homo forms radical. ✓
   Hetero forms ion. ✓
4. Addition ✓, substitution ✓, elimination ✓.
5. $Ca + Cl_2 \rightarrow Ca{-}Cl_2$ ✓

$\frac{1.203}{40.1} = 0.03$ $\quad \frac{2.13}{35.5} = 0.06$

$\frac{0.03}{0.03} = 1$ $\quad \frac{0.06}{0.03} = 2$

1:2

$CaCl_2$

Chain length

① ↑ Chain length = ↑ SA

② ↑ surface contacts between molecules

③ greater London forces between molecules

④ More energy to overcome the forces.

Branching

① fewer surface points of contact between molecules.

② branched alkanes molecules are going to be further apart

③ less London forces

④ less energy to overcome London forces.

In the presence of sunlight, alkanes can react with halogens.

UV radiation provides energy for reaction.

$$CH_{4(g)} + Br_{2(l)} \rightarrow CH_3Br_{(g)} + HBr_{(g)}$$

Bromination

Step 1 Initiation: Br-Br bonds are broken by homolytic fission.

Step 2 Propogation: Production of CH_3Br

Step 3 Termination: 2 radicals collide to stop the chain reaction.

① Initiation

$Br-Br \rightarrow \cdot Br \quad \cdot Br$

② Propogation

P1 $\cdot Br + CH_4 \rightarrow \cdot CH_3 + HBr$

P2 $\cdot CH_3 + Br_2 \rightarrow CH_3Br + \cdot Br$

Chain reaction, without termination this would be an endless cycle.

③ Termination → radicals collide.

$\cdot Br + \cdot Br \rightarrow Br_2$

$\cdot CH_3 + \cdot CH_3 \rightarrow C_2H_6$

$\cdot CH_3 + \cdot Br \rightarrow CH_3Br$

1. dodecane

$2C_{12}H_{26} + 37O_2 \rightarrow 24CO_2 + 26H_2O$

$CH_4 \xrightarrow[-HBr]{+Br_2} CH_3Br \xrightarrow[-HBr]{+Br_2} CH_2Br_2 \xrightarrow[-HBr]{+Br_2} CHBr_3 \rightarrow \cdots$

methane + ethane only form one Monosubstituted product

Longer chain alkanes form mono substituted isomers - substitution occurs at different points.

Alkenes

- unsaturated hydrocarbons
- At least 1 C=C
- Aliphatic general formula C_nH_{2n}
- Can be branched + contain multiple double bonds or be cyclic

Summarising Sigma and Pi bonds

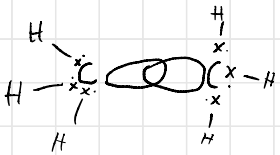
Sigma bonds:

- Covalent bonds formed from the **direct overlap** of two atomic orbitals
- Electrons are close to the nucleus of each atom = **strong bonds**

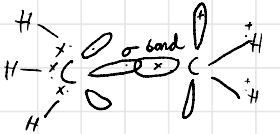
Pi bonds:

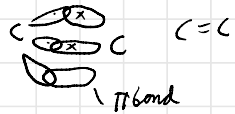
- Covalent bonds formed from the sideways **overlap** of two p-shaped atomic orbitals
- Electrons are further from the nucleus of each atom compared to sigma bond = **weaker bonds**

Ethane



Ethene





1. unsaturated = contains double bond
2. C_9H_{18} = non-4-ene

   C-C-C-C=C-C-C-C-C
3. σ = weaker,

25.04.23

1. C=C σ + π
2. C_nH_{2n}
3. Contains carbon = carbon multiple bonds.
4. Initiation, Propagation, Termination
5. 24 dm^3 101kPa

$PV = nRT$ — K°
Pa — m^3 — moles — 8.314

101,000 Pa
0.024 m^3

$T = \frac{PV}{nR}$ $\quad \frac{101{,}000 \times 0.024}{R} = 291.54$

292 K°
−273
19°C

Isomers

Structural vs Stereoisomers
Same molecular, different structural
Same structural, different arrangement in space.

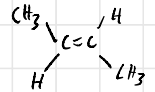
Stereoisomers

optical — E/Z

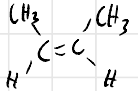
Wide range of structures

C=C only
Diff groups attached to each carbon of double bond.

E isomerism



(E) but-2-ene
entgegen (opposite)



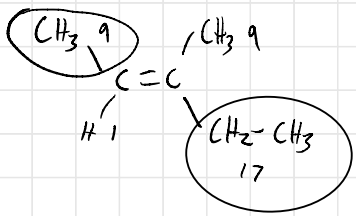
Z but-2-ene
zusammen (together)

Cis-Trans isomerism
- Special case
- each C=C, C has different
- one of attached groups must be H_2

Trans = opposite sides = E
Cis = Same sides = Z

Cahn-Ingold-Prelog (CIg) priority:

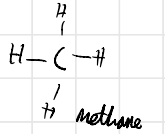
Higher Priority on Same Side = Z
Higher priority diagonally placed = E



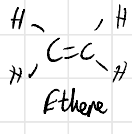
Higher Priority on same side = Z
Higher Priority diagonal = E

1. Different arrangement in space.

"Alkanes are more reactive than alkenes"



methane



Ethene

Wrong. Alkenes more reactive than alkanes because of weak π bond in the double bond. π bond weak and easier to break. Alkenes are a π and σ bond. Pi bond is exposed and above and below sigma bond.

Activity 2: Testing for Unsaturation

Bromine water can be used to test the difference between unsaturated and saturated compounds

1. Place **cyclohex**ane in 1 test tube
2. Place **cyclohex**ene in a second test tube
3. Add bromine water in both test tubes
4. Note any observations

Substance	**Observations**
cyclohexane	
cyclohexene	

16.05.23

Initiation: $Br-Br \rightarrow Br^{\cdot} + Br^{\cdot}$

propagation: $Br^{\cdot} + CH_4 \rightarrow {\cdot}CH_3 + HBr$

$\hookrightarrow {\cdot}CH_3 + Br_2 \rightarrow CH_3Br + Br^{\cdot}$

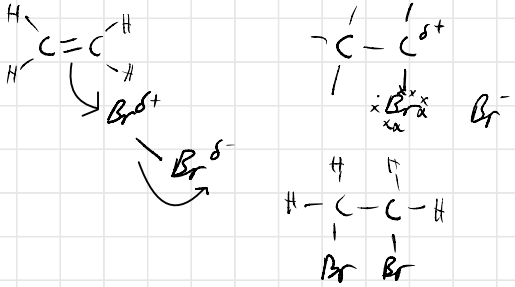
Termination $\rightarrow {\cdot}Br + Br \rightarrow Br_2$

$\rightarrow {\cdot}CH_3 + {\cdot}CH_3 \rightarrow C_2H_6$

$\rightarrow {\cdot}Br + {\cdot}CH_3 \rightarrow CH_3Br$

2. Electrophilic addition → Alkenes

Alkene → halogen Alkane



Alcohols

· C_nH_{2n+1}

methanol = fuel, Chemical feedstock.

Ethanol = Alcoholic drinks, fuel, solvent, feedstock.

physical properties of Alcohols

↑ Solubility

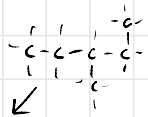
↑ volatility

↑ boiling point

Alkanes	Alcohols
non-Polar	Polar
weak london forces	weak london forces
	Stronger H bonds

19.05.23

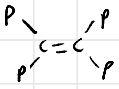
1. Cations = heterolytic fission
2. electrophile = needs an e^- (Attracted to electron rich area)
3. C-C-C-C-C-C



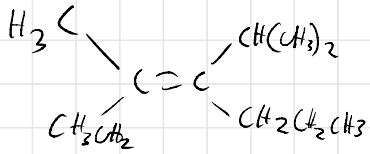
Fewer surface contact points, weaker and less London forces. Branches get in the way.

4. True
5. Primary = aldehyde, low heat
   Carboxylic acid, reflux

   Secondary = ketone

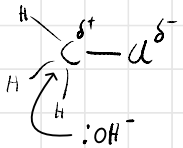


bent electron



Haloalkane = $C_nH_{2n+1}X$

Nucleophile = negative ion



Group 2 + Group 7

Group 7

less reactive:
as go down
↓ nuclear attraction
↑ shielding
↑ atomic radius

$Cl_2 + 2KBr \rightarrow 2KCl + Br_2$

Group 2

more reactive as go down
↑ nuclear attraction
↑ shielding
↑ radius

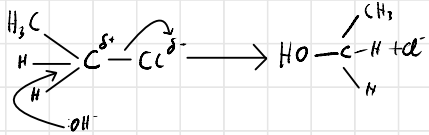
homolytic = each takes e^-
Produces radicals

heterolytic = 1 takes both pairs
+ve and -ve ion produced

Haloalkanes → nucleophilic substitution (hydrolysis)

-ve ion
lone e^- donate

electrophile:
+ve
accepts pair of e^-

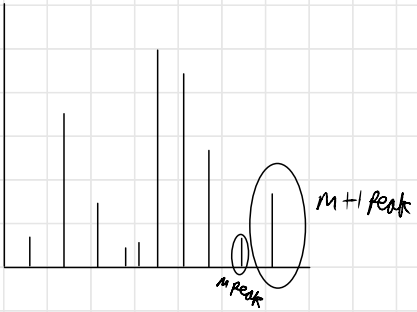


C–Cl C–Br C–I

Rate of hydrolysis increases
Carbon-halogen bond weakens

06.06.23

1. relative isotopic mass = mass of an isotope relative to carbon 12



$CH_3CH_2CH_2OH \rightarrow CH_3CH_2CH_2OH^+ + e^-$

m/z = 60

Infrared radiation + bonds